11 A car is travelling along a stretch of road at a steady speed of $11\,\text{m s}^{-1}$.

The driver accelerates, and t seconds after starting to accelerate the speed of the car, V, is modelled by the formula

$V = A + B(1 - \text{e}^{-0.17t})$.

When $t = 3$, $V = 13.8$.

(a) Find the values of A and B, giving your answers correct to 2 significant figures. **[3]**

When $t = 4$, $V = 14.5$ and when $t = 5$, $V = 14.9$.

(b) Determine whether the model is a good fit for these data. **[2]**

(c) Determine the acceleration of the car according to the model when $t = 5$, giving your answer correct to 3 decimal places. **[2]**

The car continues to accelerate until it reaches its maximum speed.

The speed limit on this road is $60\,\text{km h}^{-1}$. All drivers who exceed this speed limit are recorded by a speed camera and automatically fined £100.

(d) Determine whether, according to the model, the driver of this car is fined £100. **[3]**

11(a)

11(b)

11(c)

11(d)

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